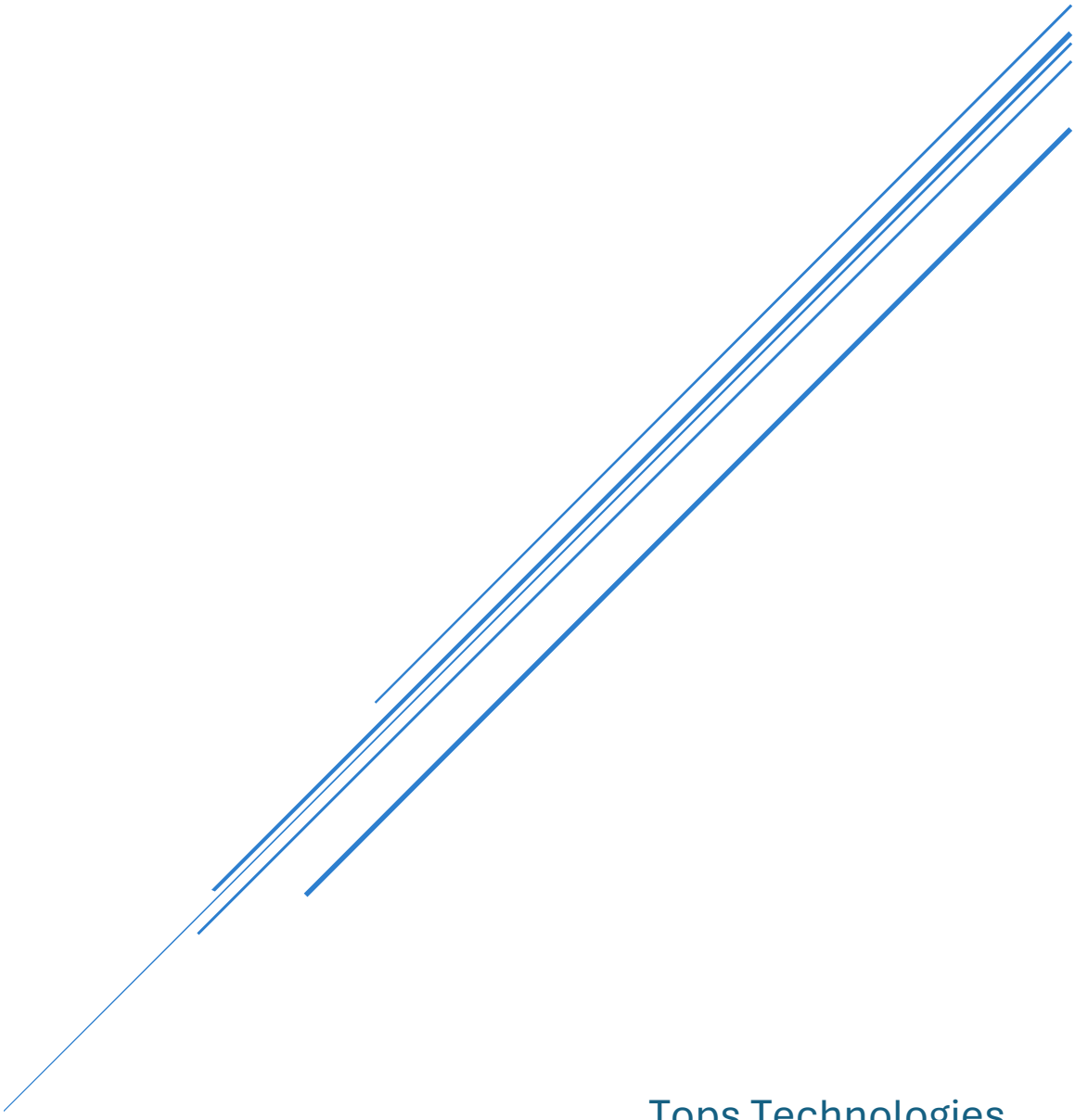
DAY – 4 HANDS-ON SUBNETTING PRACTICE

Assignment - 15



Tops Technologies
Cyber Security

1. **Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.**

➢

Given Information

- **Network Address: 192.168.10.0/24**
- **Required Subnets: 4 equal subnets**

Step 1: Calculate the new subnet mask

To create 4 subnets, we need 2 borrowed bits because \(2^2 = 4\).

- **Original CIDR: /24**
- **New CIDR: /24 + 2 = /26**
- **New Subnet Mask: 255.255.255.192**

Each subnet contains 64 total IP addresses, with 62 usable host addresses.

Step 2: Calculate the 4 subnets

Block size = \(256 - 192 = 64\).

Subnet	Network Address	Subnet Mask	Broadcast Address
1	192.168.10.0/26	255.255.255.192	192.168.10.63
2	192.168.10.64/26	255.255.255.192	192.168.10.127
3	192.168.10.128/26	255.255.255.192	192.168.10.191
4	192.168.10.192/26	255.255.255.192	192.168.10.255

Final Answer

- **Number of subnets: 4**
- **Subnet mask: 255.255.255.192 (/26)**
- **Total IP addresses per subnet: 64**
- **Usable host addresses per subnet: 62**

2. **For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.**
   Hint: Make sure not to use the network or broadcast address for any device.

➢

Let's use Subnet 1: 192.168.10.0/26 from the previous answer.

- **Network address: 192.168.10.0**
- **Subnet mask: 255.255.255.192**
- **Broadcast address: 192.168.10.63**
- **Valid host range: 192.168.10.1 – 192.168.10.62**

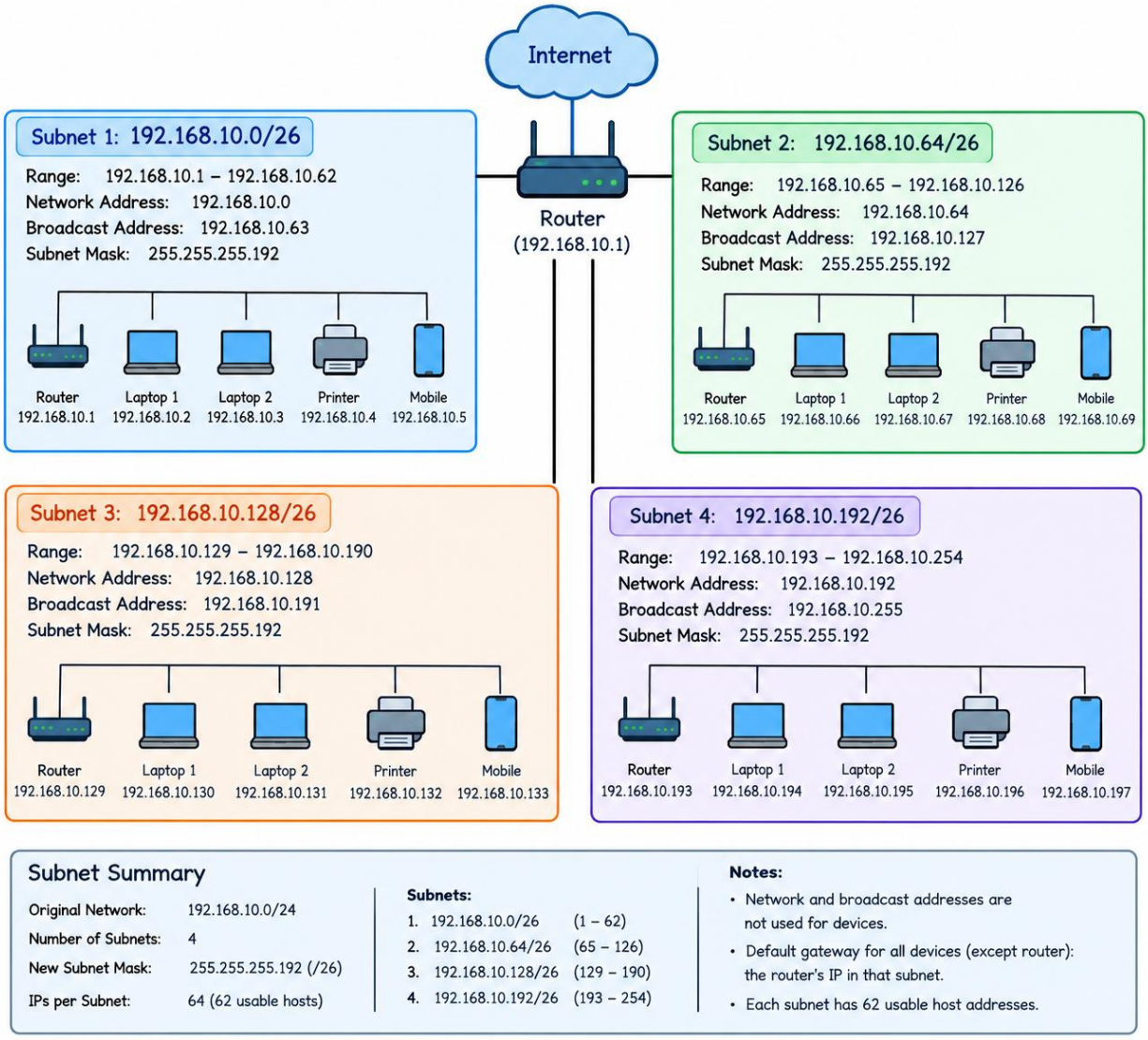
IP Address Assignment

Device	Assigned IP Address	Subnet Mask
Router	192.168.10.1	255.255.255.192
Laptop 1	192.168.10.2	255.255.255.192
Laptop 2	192.168.10.3	255.255.255.192
Printer	192.168.10.4	255.255.255.192
Mobile	192.168.10.5	255.255.255.192

Default gateway: 192.168.10.1 for all devices except the router itself.

3. **Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.**

➢



4. **Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.**

➢

Step-by-Step Subnetting Calculation

We have the network address 192.168.10.0/24 and need to divide it into 4 equal subnets. Here is how I calculated the subnet mask, hosts per subnet, and IP addresses.

Step 1: Calculate the new subnet mask

The original network is /24, which means 24 bits are used for the network portion.

To create 4 subnets, we need to borrow 2 bits from the host portion because:

$$2^2=4 \text{ subnets}$$

So, the new CIDR prefix is:

$$/24+2=/26$$

The new subnet mask is 255.255.255.192.

Step 2: Calculate the number of hosts per subnet

IPv4 addresses contain 32 bits in total.

$$32-26=6 \text{ host bits}$$

Total IP addresses per subnet:

$$2^6=64$$

Usable host addresses:

$$64-2=62 \text{ hosts}$$

We subtract 2 because one address is reserved for the network address and one for the broadcast address.

Step 3: Calculate the block size

The last octet of the subnet mask is 192.

$$\text{Block size}=256-192=64$$

Therefore, each subnet starts 64 addresses after the previous subnet.

The four network addresses are:

- **Subnet 1: 192.168.10.0**
- **Subnet 2: 192.168.10.64**
- **Subnet 3: 192.168.10.128**
- **Subnet 4: 192.168.10.192**

Step 4: Find the network and broadcast addresses

The broadcast address is one less than the next subnet's network address.

Subnet	Network Address	First Usable IP	Last Usable IP	Broadcast Address
1	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193	192.168.10.254	192.168.10.255

All four subnets use the subnet mask 255.255.255.192 (/26).

Step 5: Assign IP addresses to devices

I selected Subnet 1, 192.168.10.0/26, to connect five devices. Its valid host range is 192.168.10.1 to 192.168.10.62.

Device	Assigned IP Address	Default Gateway
Router	192.168.10.1	Not applicable
Laptop 1	192.168.10.2	192.168.10.1
Laptop 2	192.168.10.3	192.168.10.1
Printer	192.168.10.4	192.168.10.1
Mobile	192.168.10.5	192.168.10.1

I assigned a different IP address to each device to avoid IP conflicts. The router's address is the default gateway for the other devices in this subnet.

Step 6: Final summary

- **Original network: 192.168.10.0/24**

- **Number of subnets: 4**
- **New subnet mask: 255.255.255.192 (/26)**
- **Total IP addresses per subnet: 64**
- **Usable hosts per subnet: 62**
- **IP addresses assigned to devices: 5 in Subnet 1**